

Man-in-the-Middle Attack

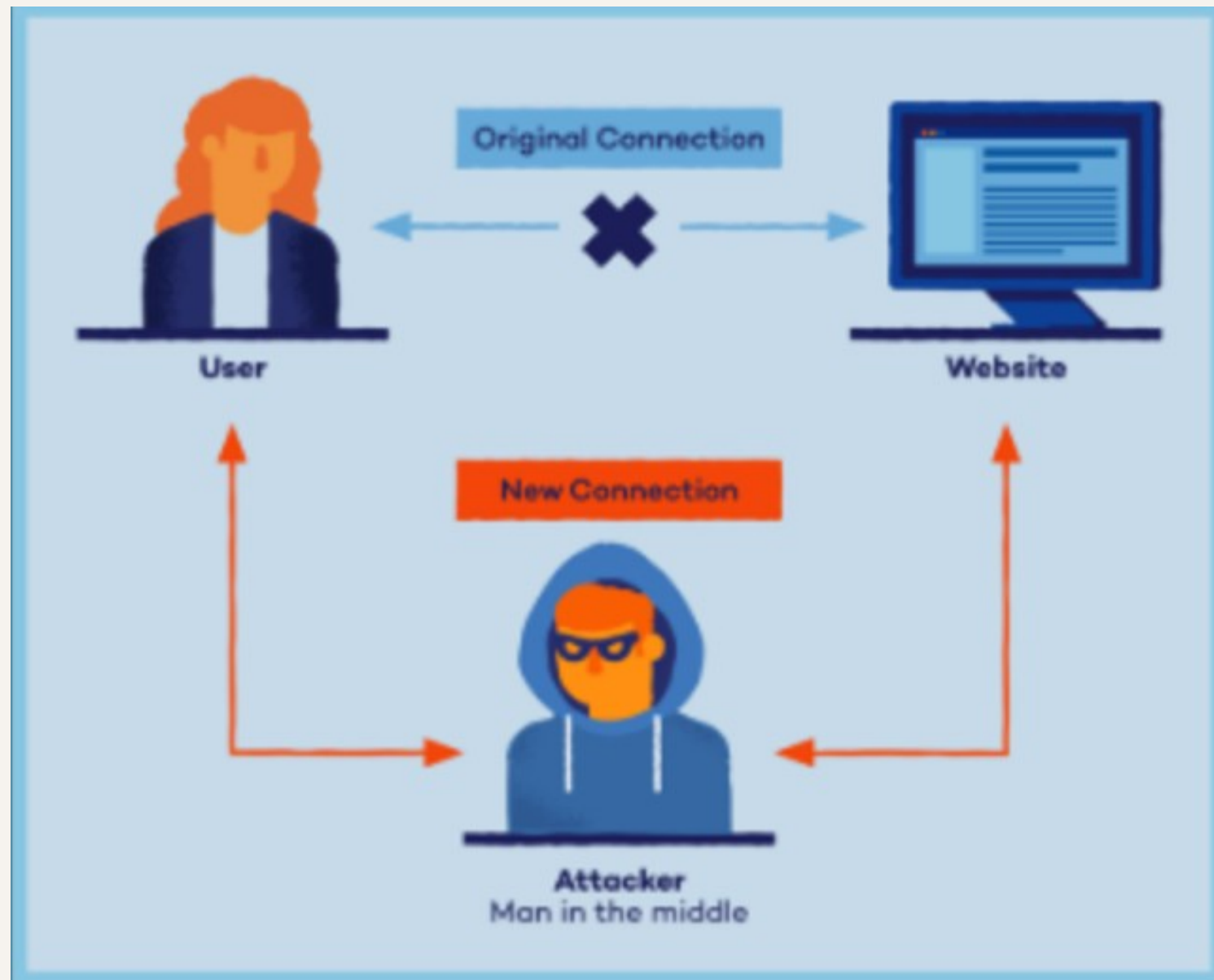
MITM

What is a Man-in-the-Middle attack?

An attacker positions themselves between two communicating parties to intercept and/or alter the data they exchange while making the communication appear normal to both.

DEFINITION

Traffic is redirected through the attacker



BEFORE

User and website talk directly

AFTER

Each side talks to the attacker, who relays the traffic

Once in the middle, the attacker may...

01

Eavesdrop

Observe and read the traffic

02

Modify

Change data before forwarding it

03

Inject

Add information

04

Drop

Stop selected information from reaching the other side

Techniques associated with MITM attacks

ARP spoofing

The attacker sends false ARP information so that a legitimate IP address is associated with the attacker's MAC address. Local network traffic is redirected through the attacker.

LEGITIMATE ENTRY

IP gateway → MAC gateway

■ FORGED ENTRY

IP gateway → MAC attacker

DNS spoofing

The attacker manipulates DNS information so that a legitimate domain name resolves to an incorrect or attacker-controlled IP address.

LEGITIMATE ANSWER

domain → server IP

■ FORGED ANSWER

domain → attacker IP

HTTP / Web spoofing

The attacker manipulates or substitutes web content so that it appears to come from the legitimate website.

EXPECTED

page from the real site

■ RECEIVED

altered or fake page

Other techniques may also be used, depending on the network and the application.

Today's deliverables

01	Extend your architecture	Add an Attacker Machine/VM
02	Demonstrate an MITM scenario	Use an appropriate technique to show how client–server communication can be intercepted or redirected
03	Analyze with Wireshark	Capture and provide evidence of the normal communication and the attack scenario
04	Explain your attack	Answer the three questions on the next slide

Explain your attack

A

How did the attacker become involved in the communication?

B

What information could the attacker observe?

C

What changed compared with normal client–server communication?

Good luck.